Department of Computer Science and Business Systems
Academic Year 2022– 2023(Odd Semester)

Degree, Semester & Branch: B.Tech – I Semester CSBS

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member (s): Dr.S.Erana Veerappa Dinesh, AP/CSE

Innovative Practice Description

Unit/Topic: Unit III / Illustrative program: Sum of an array of numbers

Course Outcome: CO3

Topic Learning Outcome: TLO10

Activity Chosen: Think Pair Share

- **Justification:**
  - ✓ Think pair share activity involves the students to think individually and to share the knowledge with classmates.
  - ✓ In python, control flow helps to help the order of an execution of program based on the values and logic.
  - ✓ The student's individual solving capability can be tested and the gained ideas can be shared by think-pair-share activity
- **Time Allotted for the Activity:** 30 Minutes

- **Details of the Implementation:**
  A Collaborative, active learning strategy, in which students work on a question posed by instructor, first individually (Think), Write in a paper, then work in pairs or groups (Pair) to solve the problem, and finally share their solution with the entire class (Share).

  • **Think –** Write- Students were provided with questions on generic methods. The student has to think individually and write the answer in a paper.
  • **Pair-** After writing the answers, the students were allowed to discuss with neighbor and justify the resultant answer to each other. The pair with neighbor was shown in Fig.2
  • **Share -** Once the conclusion is made between the pair, then it is to be shared among the whole class by any of the student from the team and it is shown in the Fig 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	P10	PSO1	PSO2	PSO3
CO 5	3	3	2	1	2	2	2	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PSO1
	3	3	2	2	1
Justification for correlation	The students apply the knowledge of control flow.	The students analyze the given program and identify solutions	Students able to design solutions using the concept control flow.	The students demonstrate problem-solving skills as a team.	The students use the generic methods that can accept any type of arguments to solve problems

- **Images / Screenshot of the practice:**

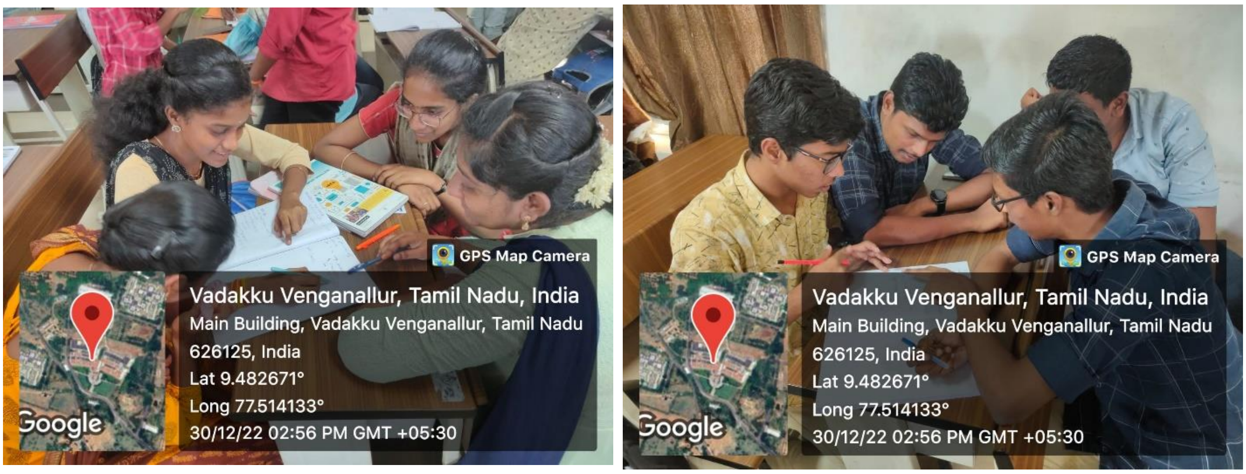

Figure 2 Pair-Sample Screen shots

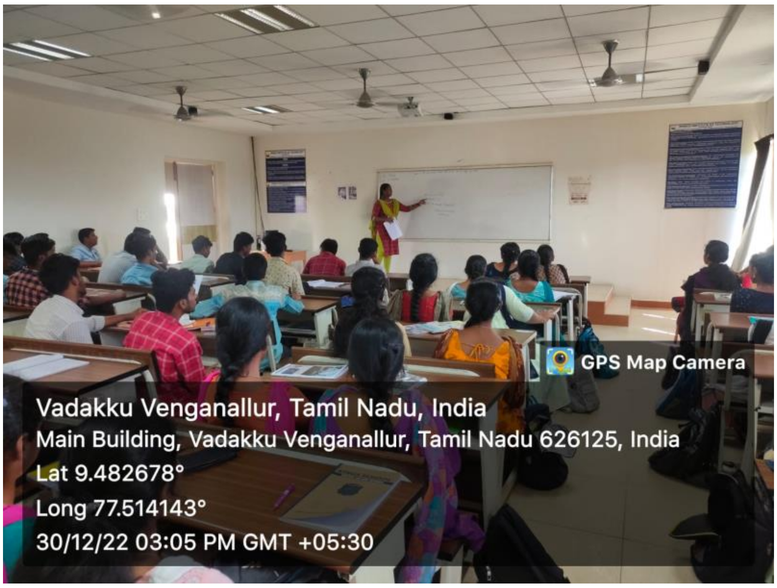

Figure 3 Share-Sample Screen shots

Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

- Some of the students write answers for the questions easily and share their ideas with others.
- Some of them need more explanation and examples to get clarity on the generic methods.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to share ideas with classmates and builds oral communication skills. They completed the task by coordinating with each other and get more confidence, clear understanding in this topic when discussing with their peers.

❖ **Challenges faced in implementation:**

- The students who didn't understand the concept clearly is not able to complete it within time.
- Making all the students to involve and present actively in this activity is a challenging.

References:

❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3_GM_CS3353_Evaluation_Expression_TPS.pdf

❖ https://www.ritrjpm.ac.in/images/computer-science/CS8451-Think-Pair-Share_PJT.pdf

❖ https://www.ritrjpm.ac.in/images/computer-science/14_CS8392_TPS.pdf

Signature of Faculty Member **HOD**